

SUNSWIFT
R A C I N G

Hydrogen
Battery
Solar

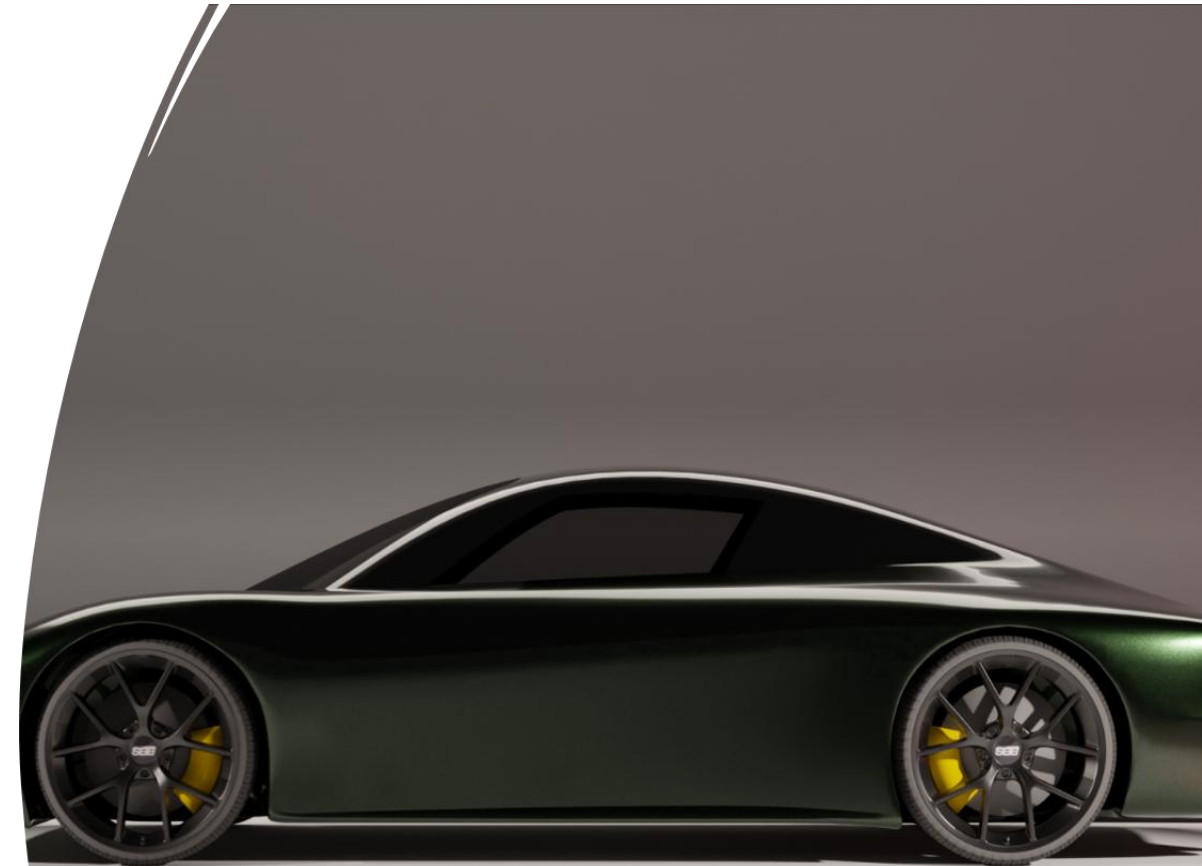
Multi-Channel Solar Car MPPT



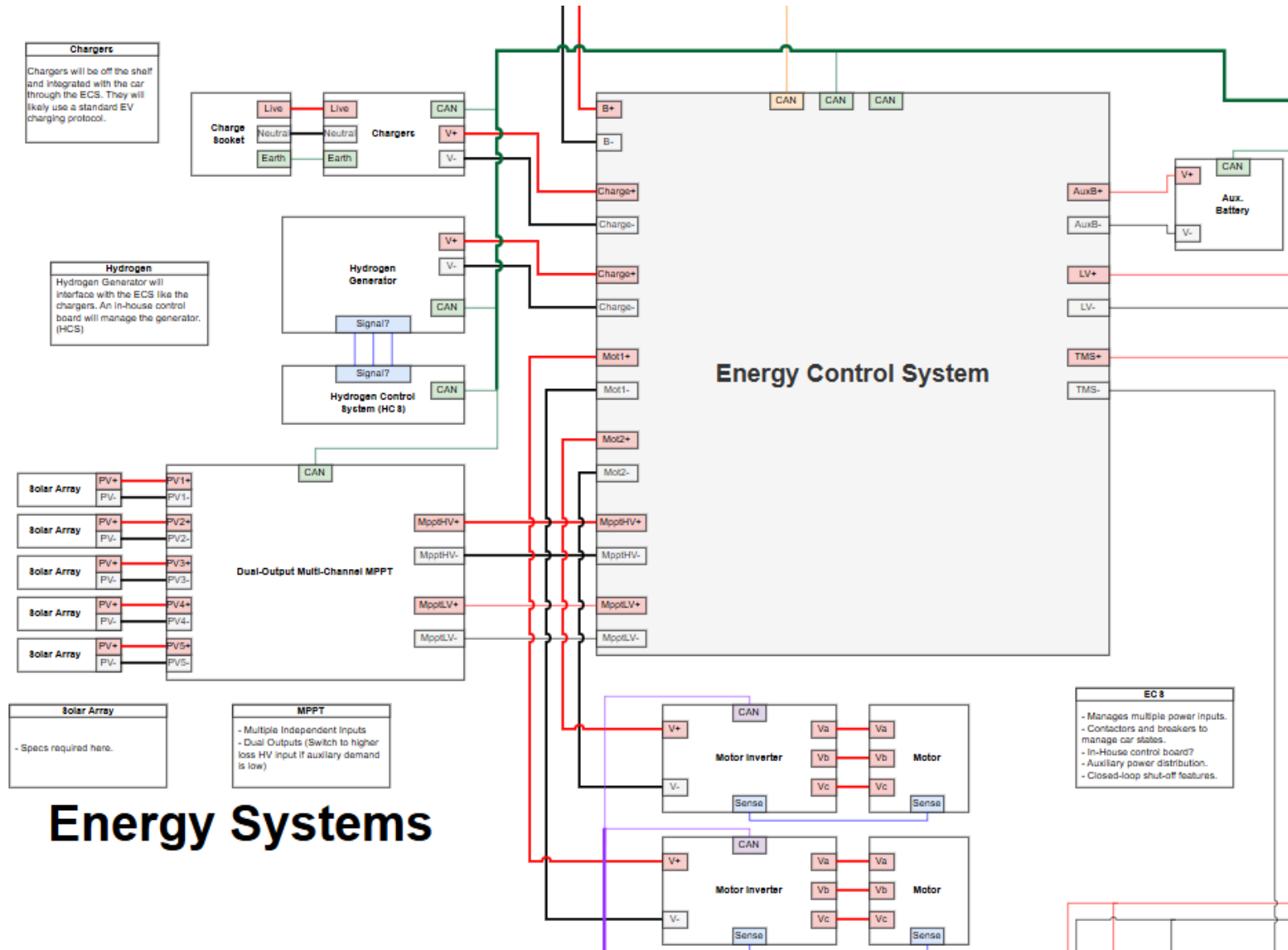
Introducing Sunswift 8

- **ADR** Compliant **TriBrid** Sports Car
 - Hydrogen
 - Solar
 - Battery

Variable	Quantity
Target Top Speed	250 km/h
Motor Power	Peak 332 kW
Battery Capacity	50 kWh
Pack Voltage	450 V – 756 V
Complete Date	December 1 st , 2026



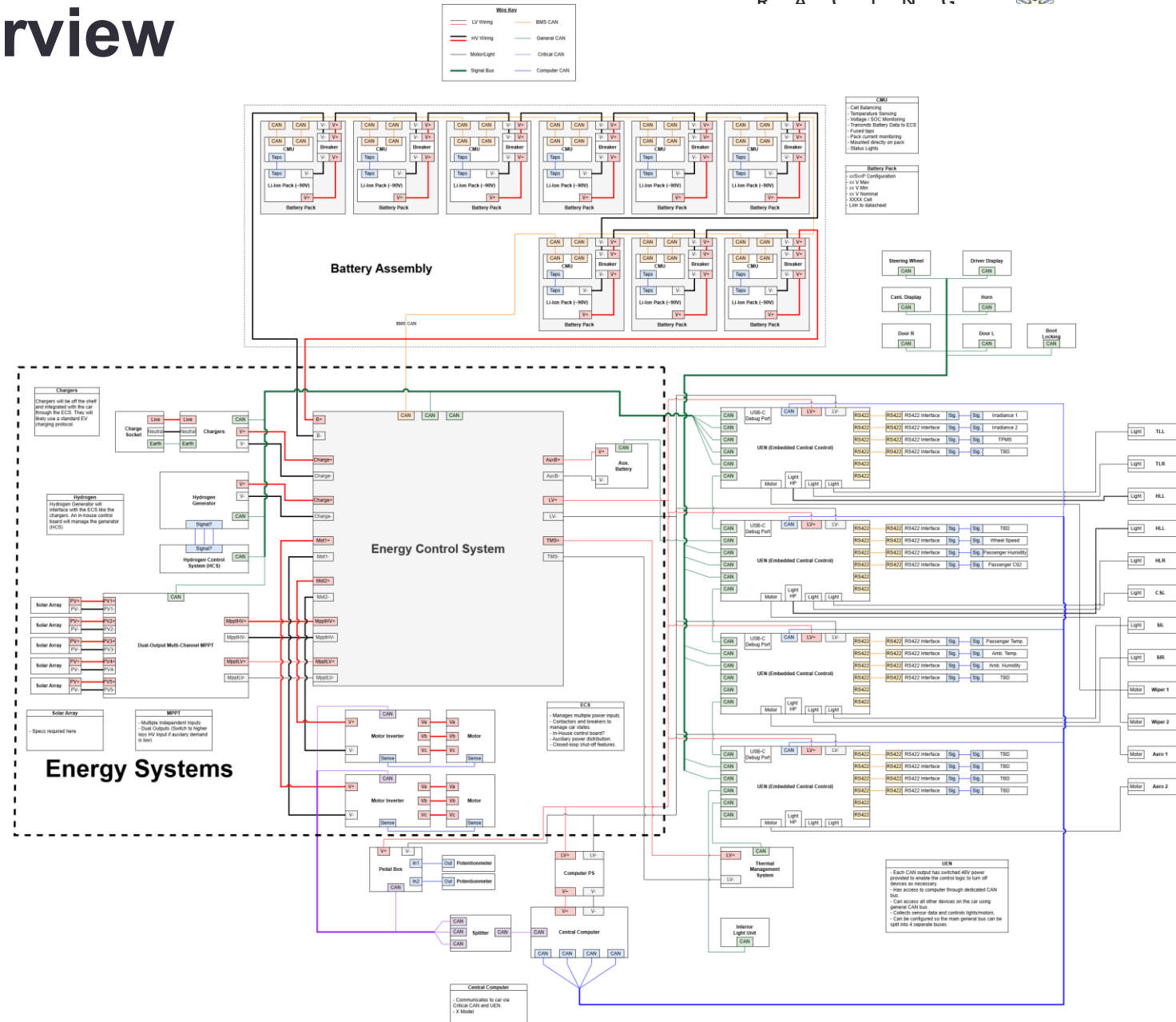
Sunswift 8 Electrical Overview



Energy Systems

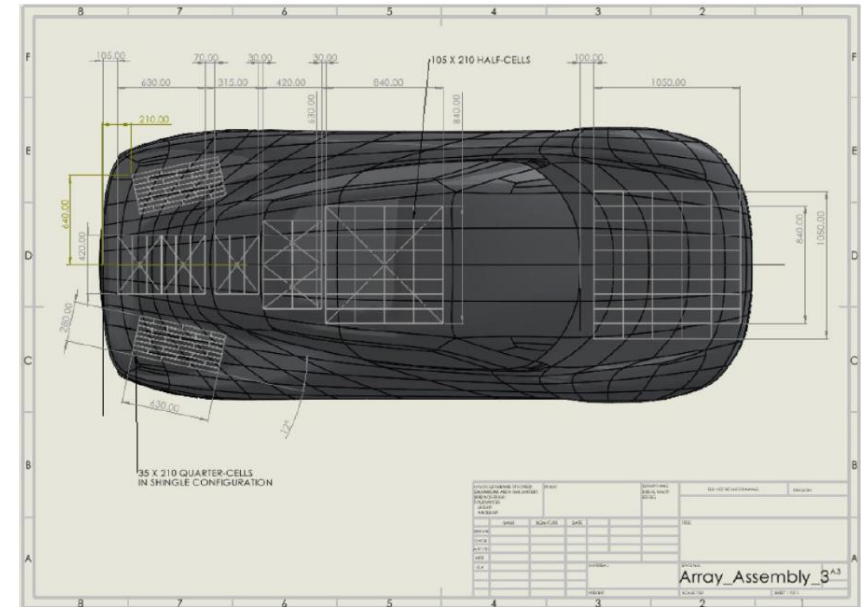
Sunswift 8 Electrical Overview

- High-level **electrical diagram** of SR8
 - Battery
 - Energy Systems
 - Embedded network
- Two primary **electrical bus's**
 - LV Bus**
 - 13S MoliceL INR-21700-P45B
 - 32.5 V minimum
 - 46.8 V nominal
 - 54.6 V maximum
 - HV Bus**
 - 180S MoliceL INR-21700-P50B
 - 450 V minimum
 - 690 V nominal
 - 756 V maximum
- Two onboard **power generation** systems
 - Hydrogen Fuel Cell
 - Solar Array



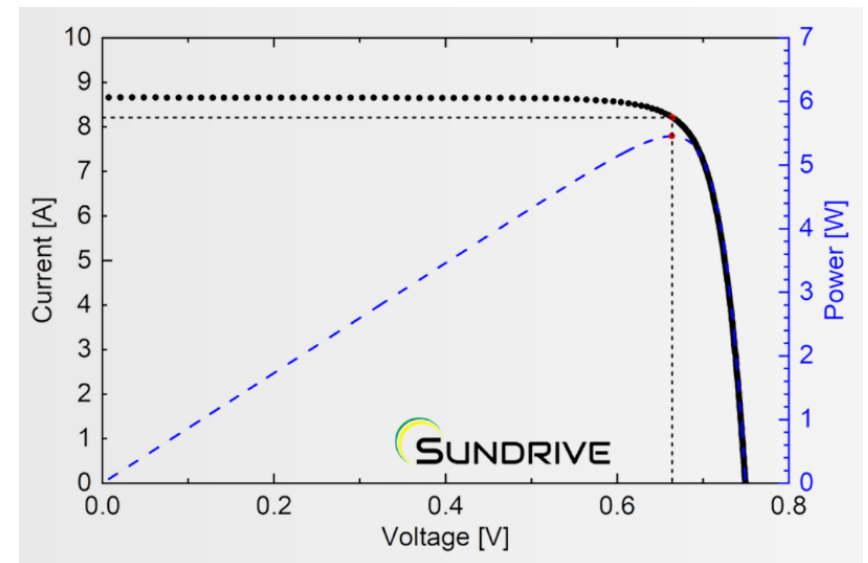
Sunswift 8 Electrical Overview - Solar

- SunDrive **G12 – 210 SD-HJT** Cells
 - 25.1% efficient
 - 19.35% power loss at 100 °C
 - 4.13% current increase at 100 °C
- **5 arrays** (all series cells)
 - All on different planes.
 - Different cell numbers.
 - Total power of 769 W.



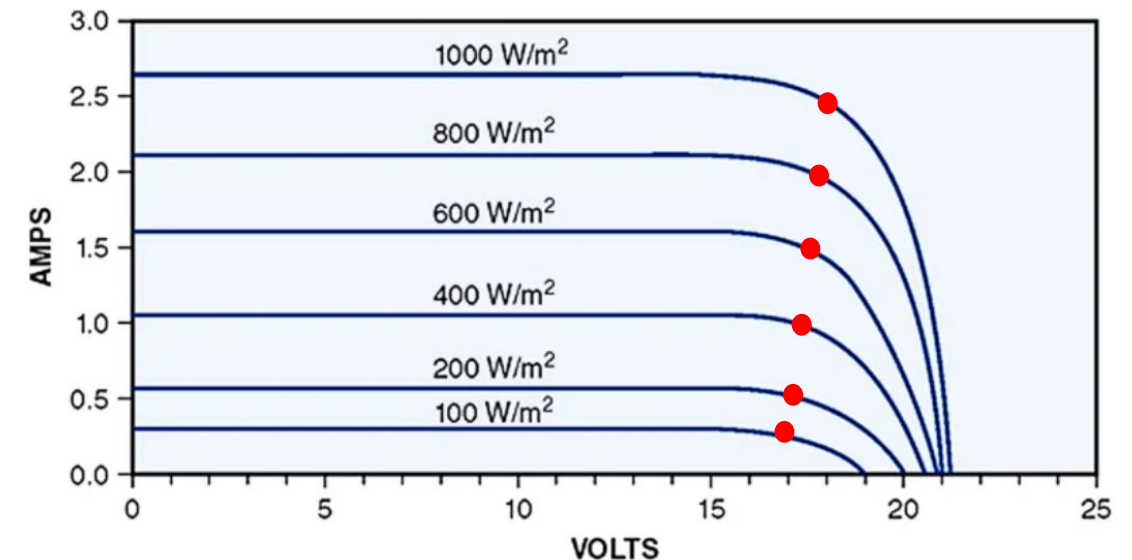
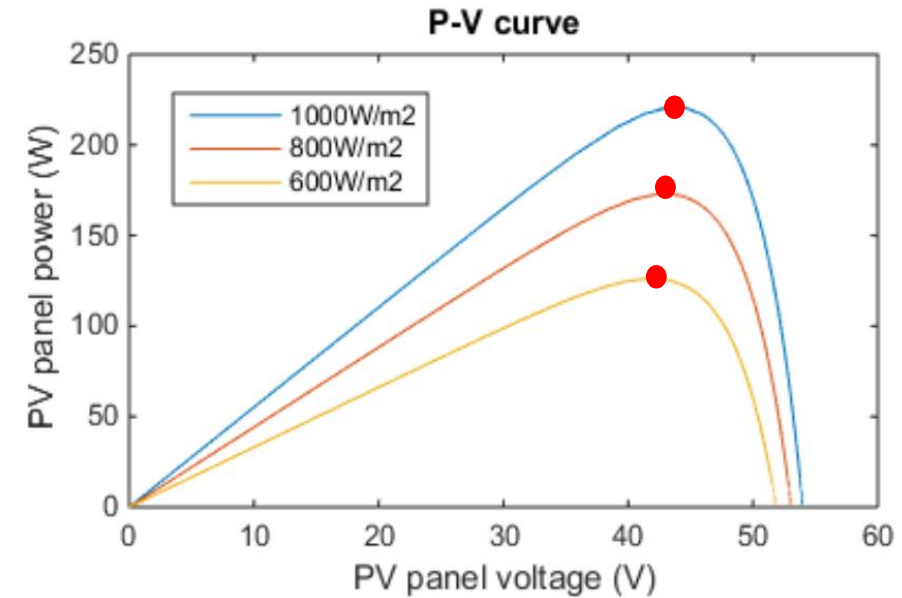
Array	Cells	V_{OC} [V]	I_{SC} [V]	V_{MPP} [V]	I_{MPP} [V]	P_{MPP} [W]
F1	41	30.50	8.66	27.59	8.22	226.81
BL	24	17.86	8.66	16.15	8.22	132.77
BR	24	17.86	8.66	16.15	8.22	132.77
R1	32	23.81	8.66	21.54	8.22	177.03
R2	18	13.39	8.66	12.11	8.22	99.58

Values calculated at STC conditions (25 °C, 1000 $\frac{W}{m^2}$, AM1.5 Spectrum)



Maximum Power Point Tracker – IV and PV Curves

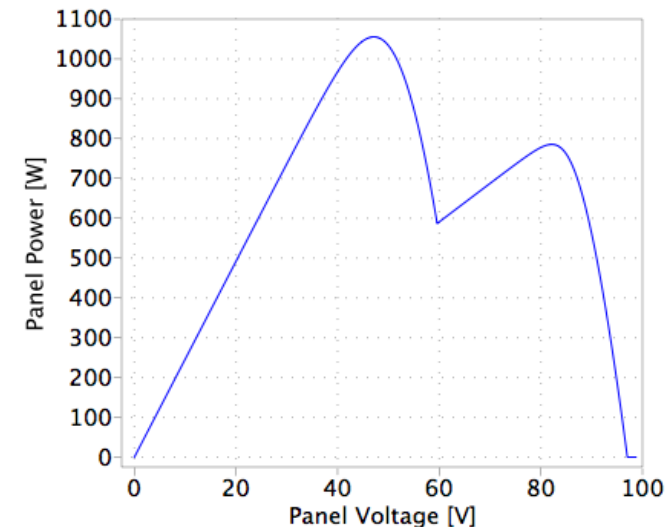
- Solar Arrays are characterized by their **IV curve**.
 - I_{SC} (y-intercept)
 - V_{OC} (x-intercept)
- The **operating point** determines how much **power is extracted**.
- P_{MPP} indicates the operating point with **maximum power extraction**.
- P_{MPP} occurs somewhere in $V \in [0, V_{OC}]$.
 - $P_{MPP} = I_{MPP}V_{MPP}$.
 - P_{MPP} occurs when $\frac{dP}{dV} = 0$.



Maximum Power Point Tracker – Purpose

- The goal is to operate the panel at P_{MPP} .
 - Not doing this results in **underutilization** of solar resources.
- To operate at P_{MPP} , it must first be found.
 - PV curves can have **multiple maxima**.
- P_{MPP} is **dynamic**. It requires active tracking.
 - Environmental variables are constantly altering the IV curve.
- P_{MPP} **changes rapidly** due to dynamic **vehicle conditions**.
 - Irradiance angle shifts as car turns.
 - Shading as car travels beneath an object.

Variable	Effect
Irradiance [G]	$G \uparrow, P_{MPP} \uparrow$
Temperature [T]	$T \uparrow, P_{MPP} \downarrow$
Cell Age [t]	$t \uparrow, P_{MPP} \downarrow$
Irradiance Angle [θ]	$\theta \uparrow, P_{MPP} \downarrow, [G_{eff} = G \cos \theta]$
Spectrum Shape	Shifts P_{MPP}
Partial Shading	Multiple P_{MPP} maxima



Maximum Power Point Tracker – Algorithm

- Algorithms involve measuring input **current** and/or **voltage**.
- Algorithm choice heavily impacts average power.
 - Each algorithm has pros/cons.
 - Flexibility is key** for efficiency.
- My thesis is dedicated to making a platform that can run almost **any MPPT algorithm**.
 - It is important that we have the necessary hardware support.
- More complex algorithms include:
 - Particle Swarm Optimization (PSO)
 - Genetic Algorithms (GA)

Common MPPT Algorithms

Algorithm	Approach
Perturb and Observe	Perturbs V_{arr} and observes P_{arr} .
Incremental Conductance	P_{MPP} occurs when $\frac{dI}{dV} = -\frac{I}{V}$.
Current Sweep	Sweeps operating point to estimate global MPP.
Temperature Method	Estimates V_{MPP} using array temperature.

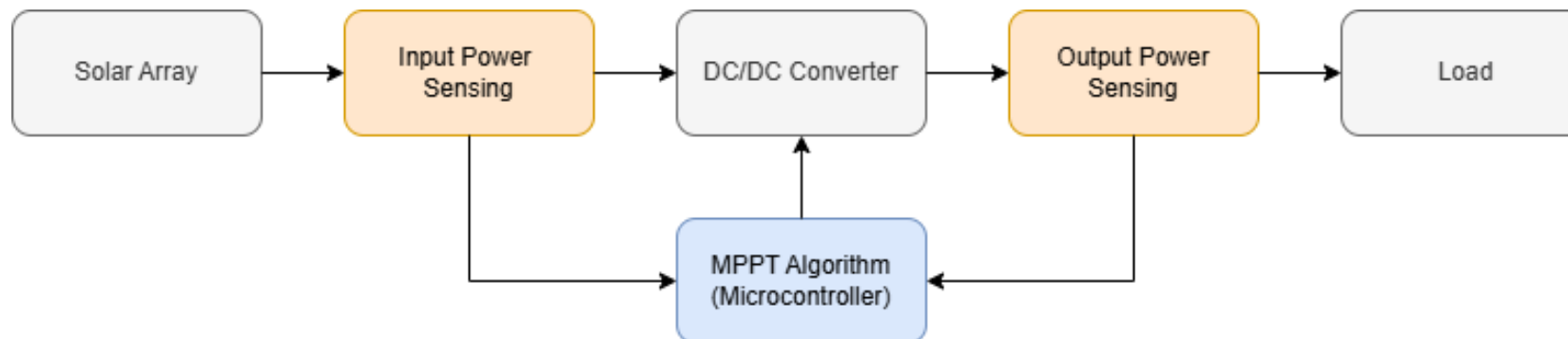
Maximum Power Point Tracker – Hardware

Auxiliary Components

- Input Power Sensing
 - To determine solar array **operating point**.
- Output Power Sensing
 - Ensures load is receiving **correct voltage**.
- Microcontroller
 - Allows feedback from power sensors.
 - **Varies DC/DC duty cycle** to load solar array at V_{MPP} .

DC/DC Converter

- Converts solar array DC power to a **useful output voltage**.
- DC/DC topology depends on **system requirements**.
 - Input and output voltage.
 - Isolation requirements.
- Some examples are:
 - Boost, Buck-Boost, Buck (Non-Isolated)
 - Flyback, Forward (Isolated)



Boost Converter Basics

Design Consideration	System Impact	Design Choice (SR8 Context)
Switching Frequency	Switching losses, inductor size/ripple current, capacitance/ripple voltage.	500 kHz to reduce inductance and maintain low ΔI .
Ripple Current	Minimum I_{in} before discontinuous conduction mode (DCM). I_{in} measurement accuracy.	$\Delta I \approx 10\% \cdot I_{in}$, since high f_{sw} . Allows for $\sim 30\mu H$ inductor.
Ripple Voltage	Output stability.	$\Delta V \approx 2\% \cdot V_{out}$.
Deadtime	Body diode loss.	$t = 5ns$, based on chosen FET datasheet recommendations.
Sync/Async	Simplicity vs efficiency.	Synchronous
Field Effect Transistor (FET) Technology	Switching frequency, conducted loss, gate drive complexity.	GaN for low $R_{DS(on)}$ and low gate charge.

Key

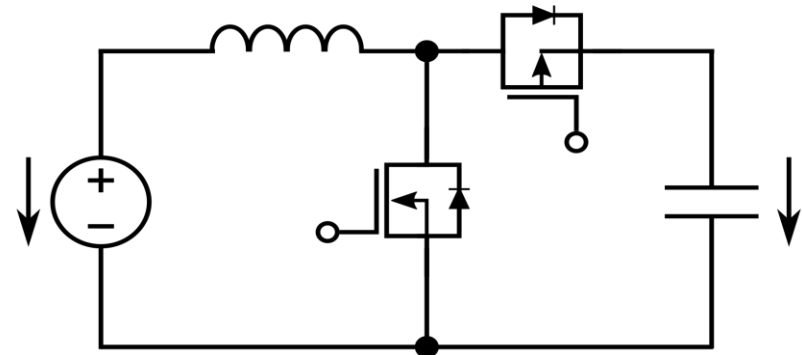
D : Duty Cycle
 t_{on} : FET on-time
 t_{off} : FET off-time
 ΔV : Ripple Voltage
 ΔI : Ripple Current
 f_{sw} : Switching Freq.

Key Equations

$$L = \frac{V_{in} \cdot D}{\Delta I \cdot f_{sw}} \quad C = \frac{I_{out}(1 - D)}{\Delta V \cdot f_{sw}}$$

$$V_{out} = \frac{V_{in}}{1 - D} \quad f_{sw} = \frac{1}{t_{on} + t_{off}}$$

Synchronous Boost Converter



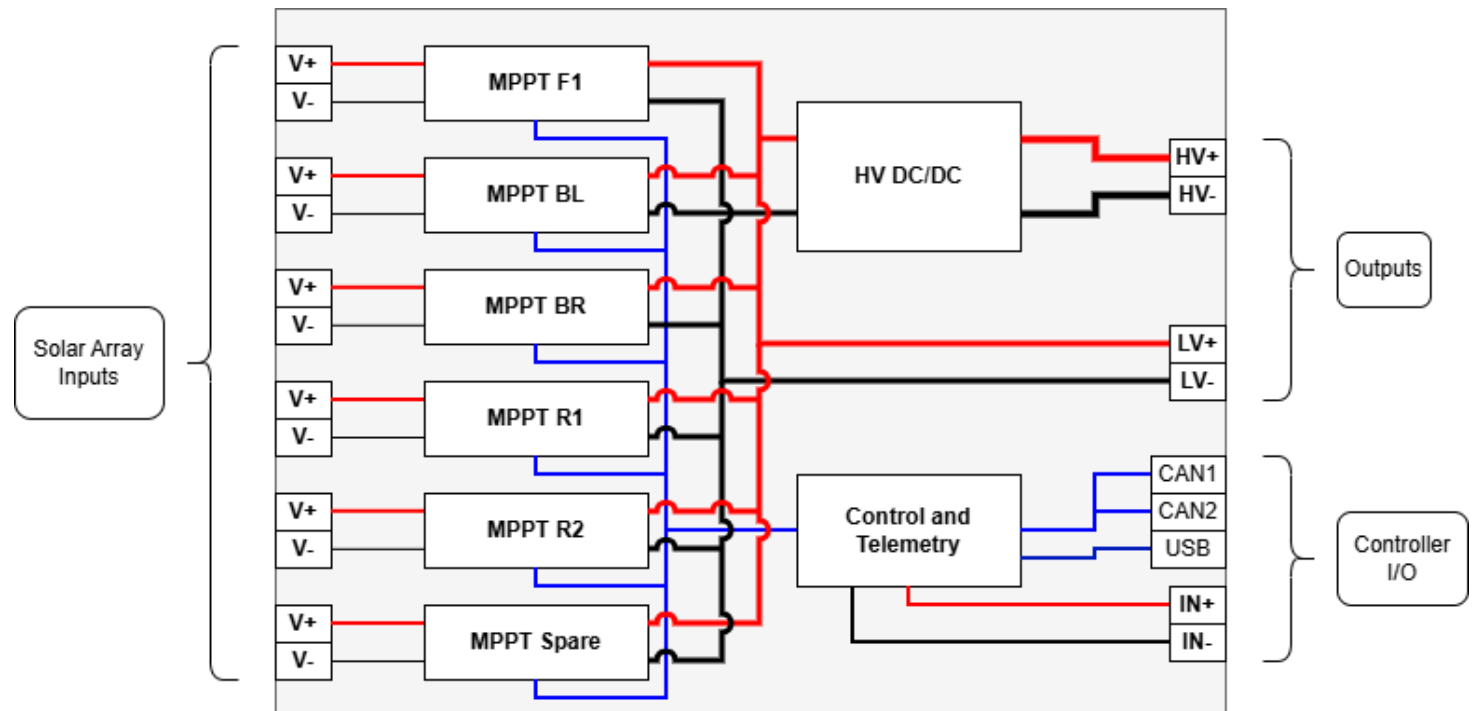
Proposed MPPT System Architecture

System Overview

- 6 **independent** MPPT channels.
 - Boost Topology
 - **Maximizes power** in mismatched conditions (shading, relative angle).
 - Optional **spare input** for V1 Hydrogen Fuel Cell (250W)
- **Dual outputs** for LV and HV bus
 - Automatically transitions to HV charging when LV battery is at capacity.
- **Centralized control architecture** using a STM32H743 with high-resolution PWM.
- Telemetry and control over FD CAN and serial.
- IP67 Enclosure for harsh operating environment.

Design Priorities

- **High-reliability** operation in a dynamic automotive environment.
- Supports a **wide variety of MPPT algorithms**.
- Designed as a **test and validation platform** for solar arrays and MPPT algorithms.



Solution Comparison

System	Pros	Cons
COTS SMPPT	• Readily available • Very high efficiency	• Require 5 due to prevent array bottlenecks • Heavy > 1 kg • Large > (20 x 10 x 10 cm) • Limited CAN integration • No algorithm control
DC Optimizers	• Per-string MPPT (reduces mismatch losses) • Proven commercial solution.	• Limited telemetry. • Not intended for automotive use. • No algorithm control. • Integration complexity (discrete modules)
My Solution	• Centralized DC Optimizer • Handles LV and HV charging • Algorithm flexibility • Rich CAN telemetry	• Design complexity. • Boutique solutions are often more expensive.

Renogy Rover Boost 10A



SolarEdge DC Optimizer



Supporting Research

Investigation of single and multiple MPPT structures of solar PV-system under partial shading conditions considering direct duty-cycle controller

Abdel-Raheem Youssef, Mostage M. Hefny, Ahmed Ismail M.Ali (Scientific Reports 13, Article number: 19051) (2023)

Key Takeaways

- Directly supports current approach.
- Small shading significantly reduces overall PV system efficiency.
- SMPPT systems are bottlenecked by weakest performing cell.
- Faster dynamic response.

“Partial shading of solar panels diminishes their operating efficiency and energy synthesized as it disrupts the uniform absorption of sunlight”

“An SMPPT can only track the MPP of a single PV module or string at a time”

“Even a small shading area... causes the entire system to operate at a sub-optimal MPP.”

Supporting Research

DC Optimizer Literature

Key Takeaways

Performance analysis of solar PV based DC optimizer distributed system with simplified MPPT method
(Springer, 2020)

DC optimizers enable **individual MPPT per panel** and eliminate mismatch losses.

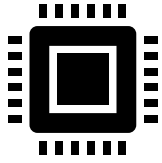
Theoretical and behavioral analysis of power optimizers for grid-connected PV systems
(Elsevier, 2022)

Power (DC) optimizers are a **distinct PV architecture** designed to handle shading, mismatch, and non-uniform conditions.

A Comprehensive Review of Distributed MPPT for Grid-Tied PV Systems at the Sub-Module Level
(Alharbi, Darwish, Ma, 2023)

Distributed MPPT systems give **better efficiency** and under shading. **better performance**

Design Challenges



Hardware

- HVDC PCB Design (~750V).
 - Creepage, clearance, insulation.
- High-efficiency, high-frequency DC/DC design.
 - Layout sensitivity due to fast GaN switching.
- Mechanical integration.
 - Thermal management.
 - Compact and waterproof.
- Robust protection and fault handling.
 - Battery protection
 - Pre-charge, overcurrent, etc.



Firmware

- Real-time control across multiple power stages.
- Software defined protection logic.
- Modular algorithm framework.
- FD CAN and serial telemetry.
- User interface for testing and deploying algorithms.



Testing & Validation

- Testing under dynamic real-world solar car conditions.
- Protection systems validation.
- Testing a HV system.
 - Dangerous and requires appropriate training.
- Vehicle integration.

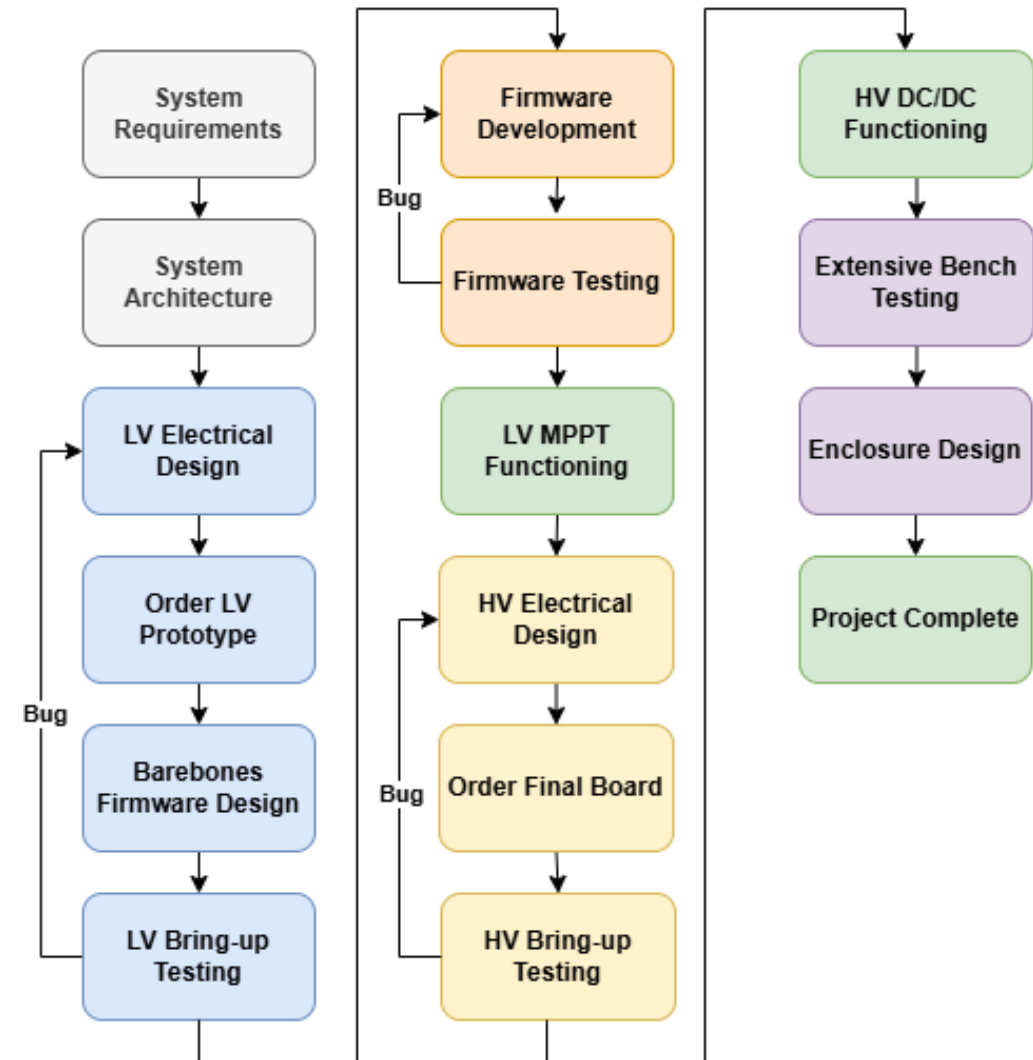
Project Plan

Project Stages

- 1 System Requirements and Architecture
- 2 LV Design and Test
- 3 Firmware Design and Test
- 4 HV Design and Test
- 5 Performance Evaluation and Integration

MVP – Achieve functioning 5-channel LV MPPT with >90% efficiency between 0-8.2 A (input).

External HV DC/DC can be procured for HV solar charging.



Required Skills and Training

Skill	Application	Proficiency
Altium Layout and Schematics	Electrical hardware development.	Strong – Designed PCBs for Sunswift and industry.
HV Design and Handling	HV DC/DC development.	Developing
Power Stage Design	Boost converter design.	Strong – Developed synchronous buck converters.
Firmware Design in C	Control and algorithm execution.	Developing
SolidWorks	Enclosure design.	Strong – Developed and printed test jigs.

Preliminary Results and Progress

Key Component Selection

- Microcontroller
 - STM32H743ZIT6
- LV GaN FET
 - LMG2100R026
 - Integrated gate driver.
- Python scripts for minimum inductance and capacitance.

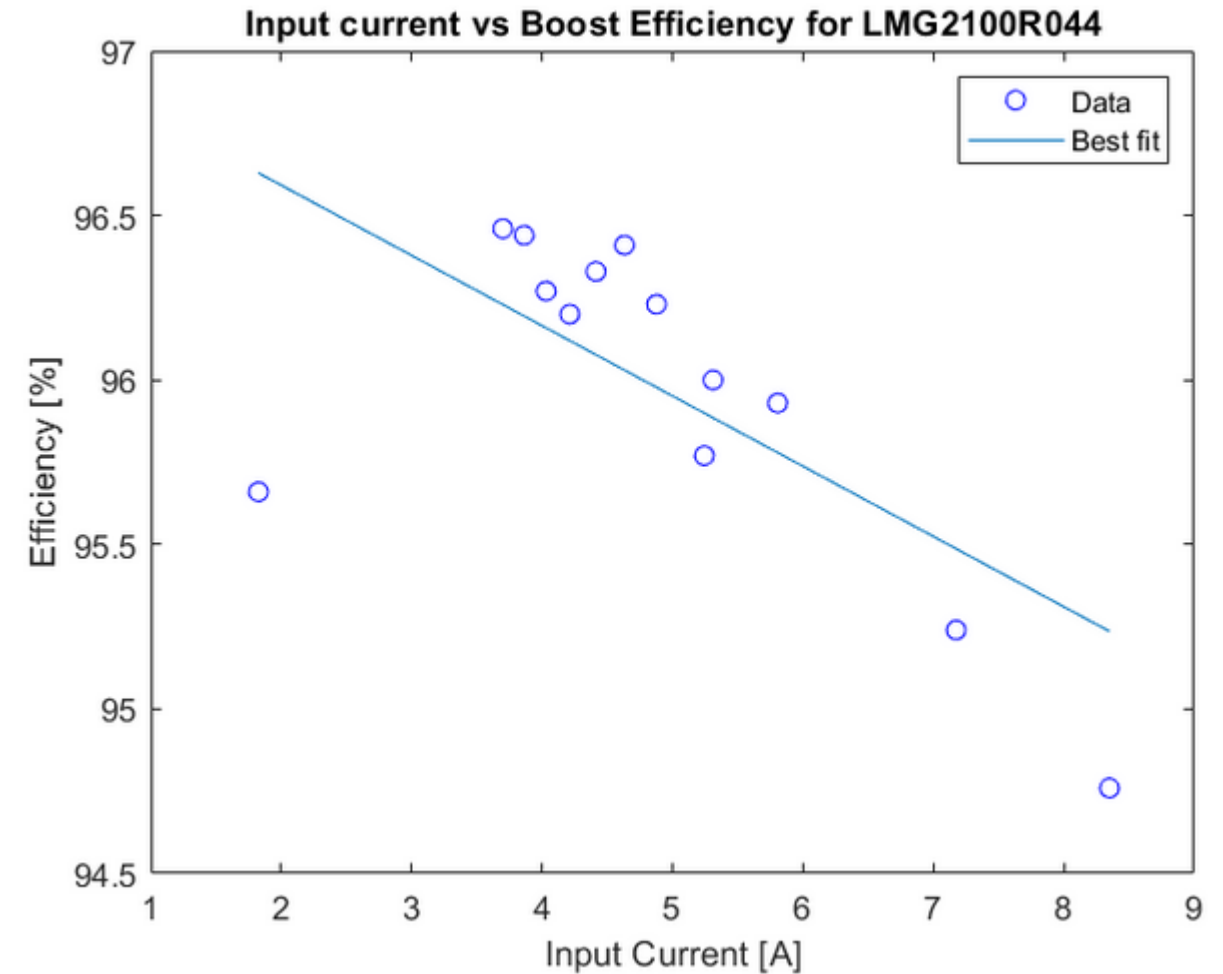
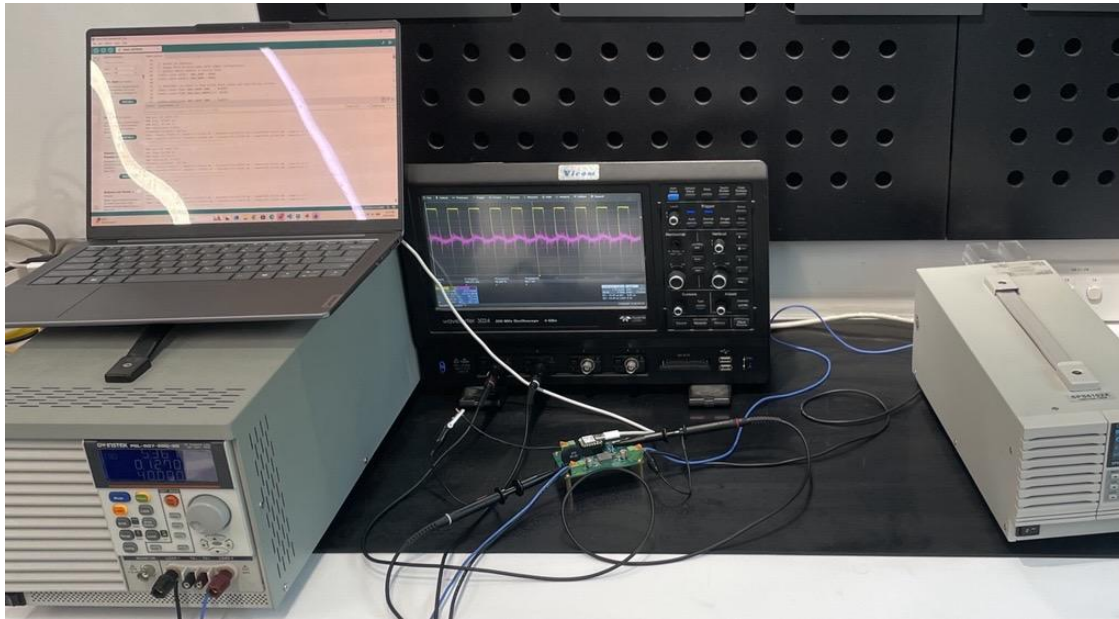
Solar Array	Inductance [μH]	Capacitance [μF]
F1	31.9	18.4
BL	26.6	6.3
BR	26.6	6.3
R1	30.5	3.5
R2	22.0	3.2

LMG2100R044 Testing

- Development board in boost configuration.
 - $22\mu H$ inductor.
 - $\sim 7\mu F$ capacitance.
- Preliminary testing showed $\sim 96\%$ efficiency at F1 voltages.
- Further testing is required; measurement inaccuracies could be up to $\pm 2\%$.
- Results gave me the confidence to select this FET and gate driver package.

Preliminary Results and Progress

Bench Testing Setup



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RACING

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Battery
Solar

Questions?



LMG2100R044 100V, 35A GaN Half-Bridge Power Stage

1 Features

- Integrated 4.4mΩ half-bridge GaN FETs and driver
- 90V continuous, 100V pulsed voltage rating
- Package optimized for easy PCB layout
- High slew rate switching with low ringing
- 5V external bias power supply
- Supports 3.3V and 5V input logic levels
- Gate driver capable of up to 10MHz switching
- Excellent propagation delay (33ns typical) and matching (2ns typical)
- Internal bootstrap supply voltage clamping to prevent GaN FET Overdrive
- Supply rail undervoltage for lockout protection
- Low power consumption
- Exposed top QFN package for top-side cooling
- Large GND pad for bottom-side cooling

2 Applications

- Buck, boost, buck-boost converters
- LLC converters
- Solar inverters
- Telecom and server power
- Motor drives
- Power tools
- Class-D audio amplifiers

3 Description

The LMG2100R044 device is a 90V continuous, 100V pulsed, 35A half-bridge power stage, with integrated gate-driver and enhancement-mode Gallium Nitride (GaN) FETs. The device consists of two 100V GaN FETs driven by one high-frequency 90V GaN FET driver in a half-bridge configuration.

GaN FETs provide significant advantages for power conversion as they have zero reverse recovery and very small input capacitance C_{ISS} and output capacitance C_{OSS} . All the devices are mounted on a completely bond-wire free package platform with minimized package parasitic elements. The LMG2100R044 device is available in a 5.5mm × 4.5mm × 0.89mm lead-free package and can be easily mounted on PCBs.

The TTL logic compatible inputs can support 3.3V and 5V logic levels regardless of the VCC voltage. The proprietary bootstrap voltage clamping technique ensures the gate voltages of the enhancement mode GaN FETs are within a safe operating range.

The device extends advantages of discrete GaN FETs by offering a more user-friendly interface. It is an ideal solution for applications requiring high-frequency, high-efficiency operation in a small form factor.

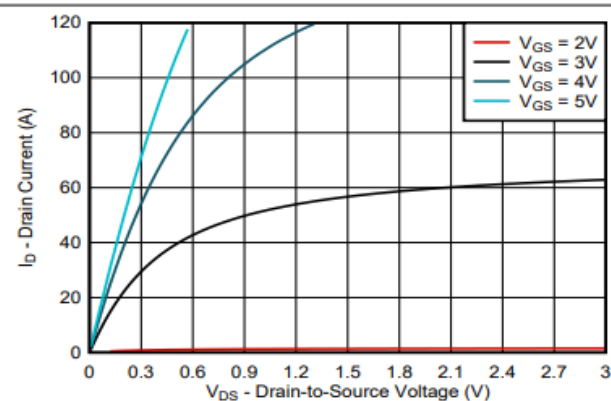


Figure 5-1. Typical Output Characteristics at 25°C

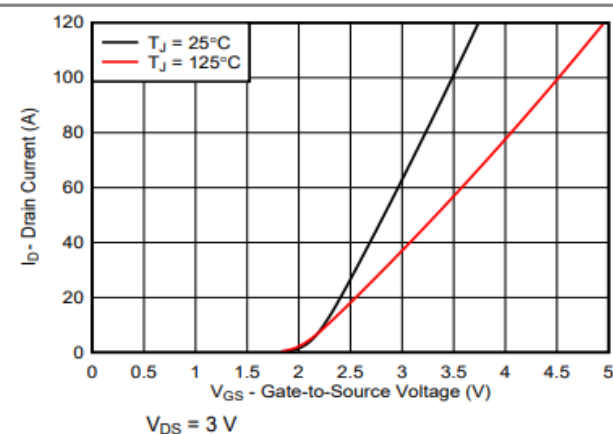


Figure 5-2. Typical Transfer Characteristics

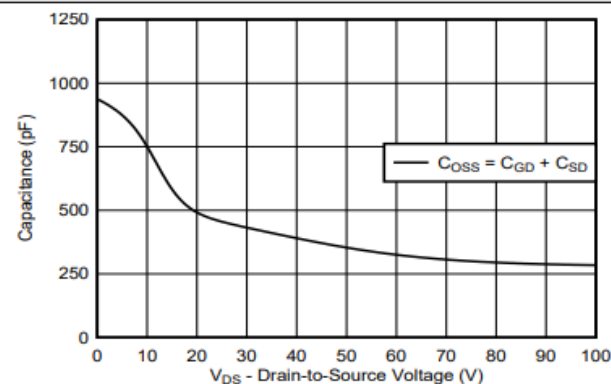


Figure 5-3. Typical Capacitance (Linear Scale)

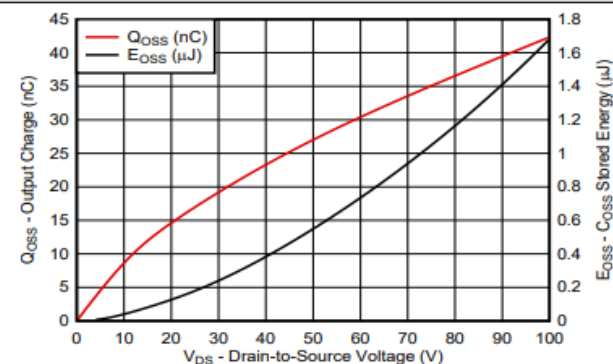


Figure 5-4. Typical Output Charge and C_{OSS} Stored Energy



Cell Type	N-type G12 SunDrive HJT
Material	Monocrystalline Silicon
Cell Dimension	210 mm x 105 mm $\pm$ 0.25
Thickness	120 μ m $\pm$ 20 μ m
Cell Efficiency	25.1 % $\pm$ 0.1 %
No. of Busbars	18

Temperature Coefficient of Pmax	-0.258 %/°C
Temperature Coefficient of Voc	-0.27 %/°C
Temperature Coefficient of Isc	0.055 %/°C

Efficiency - Eff	25.1 % $\pm$ 0.1 %
Maximum power point power - Pmpp	5.53 W
Open-circuit voltage - Voc	0.744 V
Short-circuit current - Isc	8.66 A
Fill Factor	85.3 %
Maximum power point voltage - Vmpp	0.673 V
Maximum power point current - Impp	8.22 A

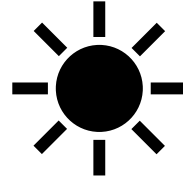
- Irradiance 1000 W/m
- Temperature 25 °C
- AM1.5G Spectrum

Why this architecture?



Solar Car Challenges

- Frequent partial shading.
- Rapid P_{MPP} changes due to dynamic conditions.
- Tight space requirements.
- Environmental constraints.
 - Vibration.
 - Waterproofing.



COTS Limitations

- **Significantly larger** when implementing equivalent functionality using COTS products.
- **No control over MPPT algorithm** implementation.
- No dual output support at our required voltages.



Solution Advantages

- Optimal per-array tracking.
- Dual LV/HV output in single package.
- Compact design enabling simplified SR8 integration.
- **Flexible platform for algorithm development** and array validation.

Enables maximum energy extraction under all operating conditions while maintaining reliability and flexibility with tight system constraints.